

Test Report - ANNEX B

การทดสอบ

วันที่ทดสอบ

Charger detail

Brand

Rated

Chargepoint ID

สรุปลผลการทดสอบ

Open Charge Point Protocol (OCPP) และการสื่อสารข้อมูลกับ MEA OCPP Server (OCPP 1.6j) : CCS2 x2 at Warehouse 10%

เวลาเริ่มทดสอบ

เวลาสิ้นสุดการทดสอบ

Model

Firmware version

NO. of Connector ID

Connector type

CCS 2

รายการทดสอบ

ลำดับ	Message	MEA OCPP Sever Required			Remark
		รายละเอียดการทดสอบ	Conn. ID 1	Conn. ID 2	
1	การตรวจสอบการตั้งค่าเครื่องชาร์จ				
1.1	BootNotification	Include : chargePointModel, chargePointVendor, firmwareVersion			
1.2	StatusNotification	Include : connectorId (All Connector ID [0,1,2,3,...]), errorCode, info, status, timestamp, vendorId, vendorErrorCode			
1.3	TriggerMessage (BootNotification)	Include : status (Accepted)			
1.4	BootNotification	Include : chargePointModel, chargePointVendor, firmwareVersion			
1.5	TriggerMessage (StatusNotification)	Include : status (Accepted)			
1.6	StatusNotification	Include : connectorId (All Connector ID [0,1,2,3,...]), errorCode, info, status, timestamp, vendorId, vendorErrorCode			
1.7	TriggerMessage (MeterValues)	Include : status (Accepted)			
1.8	MeterValues	Include : connectorId, transactionId, meterValue {timestamp, sampledValue {measurand (Voltage, Current.Import, Energy.Active.Import.Register, Power.Active.Import, SoC , Power.Offered), value, unit, context, location}}			
1.9	GetConfiguration	Include : configurationKey {HeartbeatInterval (int), LocalAuthorizeOffline (boolean), MeterValueSampleInterval (int), UnlockConnectorOnEVSideDisconnect (boolean)}			
1.10	ChangeConfiguration (HeartbeatInterval: 10 min)	Include : status (Accepted)			

รายการทดสอบ

ลำดับ	Message	MEA OCPP Sever Required			Remark
		รายละเอียดการทดสอบ	Conn. ID 1	Conn. ID 2	
1.11	ChangeConfiguration (MeterValueSampleInterval: 30 sec)	Include : status (Accepted)			
1.12	ChangeConfiguration (UnlockConnectorOnEVSideDisconnect: True)	Include : status (Accepted)			
1.13	ChangeAvailability (connectorId, Inoperative)	Include : status (Accepted)			
1.14	StatusNotification	Include : connectorId (All Connector ID [0,1,2,3,...]), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
1.15	ChangeAvailability (connectorId, Operative)	Include : status (Accepted)			
1.16	StatusNotification	Include : connectorId (All Connector ID [0,1,2,3,...]), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
1.17	GetDiagnostics	Include : fileName			*ขึ้นกับการพิจารณา
1.18	DiagnosticsStatusNotification	Include : status (Uploading, Uploaded,UploadFailed)			*ขึ้นกับการพิจารณา
1.19	UpdateFirmware	Include :			*ขึ้นกับการพิจารณา
1.20	FirmwareStatusNotification	Include : status (Downloading, Downloaded, Installing, Installed)			*ขึ้นกับการพิจารณา
1.21	ChangeConfiguration (LocalAuthorizeOffline: True)	Include : status (Accepted)			
1.22	SendLocalList (listVersion, updateType: Full , [localAuthorizationList])	Include : status (Accepted)			
1.23	GetLocalListVersion	Include : listVersion			
1.24	clearCache	Include : status (Accepted)			
2	การตรวจสอบการทำงาน Auto Charge				"Authorize", {"idTag": "VID:[id]"}
2.1	ChangeConfiguration (AutoCharge: False)	Include : status (Accepted)			*หากไม่รองรับการ ChangeConfiguration จะต้องสามารถ Config ที่หน้าจอของเครื่องอัดประจุไฟฟ้าได้
2.2	StatusNotification (Plug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			*เมื่อเสียบหัวกับรถแล้ว เครื่องอัดประจุไฟฟ้าจะต้องไม่ส่งขอ Authorize ด้วย VID
2.3	StatusNotification (Unplug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			

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ลำดับ	Message	MEA OCPP Sever Required			Remark
		รายละเอียดการทดสอบ	Conn. ID 1	Conn. ID 2	
2.4	ChangeConfiguration (AutoCharge: True)	Include : status (Accepted)			*หากไม่รองรับการ ChangeConfiguration จะต้องสามารถ Config ที่หน้าจอของเครื่องอัดประจุไฟฟ้าได้
2.5	StatusNotification (Plug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
2.6	Authorize (Auto Charge)	Include : idTag (Accepted)			{"idTag": "VID:[id]"} EX. : {"idTag": "VID:FCA47A10E365"}
2.7	StartTransaction	Include : connectorId , idTag , meterStart , timestamp			
2.8	StatusNotification	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
2.9	MeterValues (30 sec) *Energy.Active.Import.Register must be related to meterStart	Include : connectorId , transactionId , meterValue { timestamp , sampledValue { measurand (Voltage, Current.Import, Energy.Active.Import.Register, Power.Active.Import, SoC , Power.Offered), value , unit , context , location }}			
2.10	RemoteStopTransaction	Include : status			
2.11	StopTransaction	Include : idTag , meterStop , timestamp , transactionId			
2.12	StatusNotification (Plug) (If any)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
2.13	StatusNotification (Unplug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
2.14	StatusNotification (Plug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
2.15	Authorize (Auto Charge)	Include : idTag (Accepted)			{"idTag": "VID:[id]"} EX. : {"idTag": "VID:FCA47A10E365"}
2.16	StartTransaction	Include : connectorId , idTag , meterStart , timestamp			
2.17	StatusNotification	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
2.18	MeterValues (30 sec) *Energy.Active.Import.Register must be related to meterStart	Include : connectorId , transactionId , meterValue { timestamp , sampledValue { measurand (Voltage, Current.Import, Energy.Active.Import.Register, Power.Active.Import, SoC , Power.Offered), value , unit , context , location }}			

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ลำดับ	Message	MEA OCPP Sever Required			Remark
		รายละเอียดการทดสอบ	Conn. ID 1	Conn. ID 2	
2.19	StatusNotification (Suspended by EV)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode, info, status, timestamp, vendorId, vendorErrorCode			
2.20	StopTransaction (Unplug)	Include : idTag, meterStop, timestamp, transactionId			
2.21	StatusNotification (Unplug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode, info, status, timestamp, vendorId, vendorErrorCode			*หลังเสร็จสิ้นหัวข้อนี้ ผู้ดำเนินการทดสอบ จะต้องตั้งสถานะของ VID เป็น Invalid
3	การตรวจสอบการทำงานรูปแบบปกติ				
3.1	BootNotification	Include : chargePointModel, chargePointVendor, firmwareVersion,			
3.2	StatusNotification	Include : connectorId (All Connector ID [0,1,2,3,...]), errorCode, info, status, timestamp, vendorId, vendorErrorCode			
3.3	StatusNotification (Plug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode, info, status, timestamp, vendorId, vendorErrorCode			
3.4	Authorize (Valid Card)	Include : IdTag (Accepted)			
3.5	StartTransaction	Include : connectorId, idTag, meterStart, timestamp			
3.6	StatusNotification	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode, info, status, timestamp, vendorId, vendorErrorCode			
3.7	MeterValues (30 sec) *Energy.Active.Import.Register must be related to meterStart	Include : connectorId, transactionId, meterValue {timestamp, sampledValue {measurand (Voltage, Current.Import, Energy.Active.Import.Register, Power.Active.Import, SoC , Power.Offered), value, unit, context, location}}			
3.8	StopTransaction (Valid Card)	Include : idTag, meterStop, timestamp, transactionId			
3.9	StatusNotification (Plug) (If any)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode, info, status, timestamp, vendorId, vendorErrorCode			
3.10	StatusNotification (Unplug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode, info, status, timestamp, vendorId, vendorErrorCode			
3.11	StatusNotification (Plug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode, info, status, timestamp, vendorId, vendorErrorCode			
3.12	RemoteStartTransaction	Include : status (Accepted)			
3.13	StartTransaction	Include : connectorId, idTag, meterStart, timestamp			

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ลำดับ	Message	MEA OCPP Sever Required			Remark
		รายละเอียดการทดสอบ	Conn. ID 1	Conn. ID 2	
3.14	StatusNotification	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
3.15	MeterValues (30 sec) *Energy.Active.Import.Register must be related to meterStart	Include : connectorId , transactionId , meterValue { timestamp , sampledValue {measurand (Voltage, Current.Import, Energy.Active.Import.Register, Power.Active.Import, SoC , Power.Offered), value , unit, context, location}}			
3.16	RemoteStopTransaction	Include : status			
3.17	StopTransaction	Include : idTag , meterStop , timestamp , transactionId			
3.18	StatusNotification (Plug) (If any)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
3.19	StatusNotification (Unplug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
4	การตรวจสอบการรีเซ็ต				
4.1	Reset (Hard)	Include : status			
4.2	BootNotification	Include : chargePointModel , chargePointVendor , firmwareVersion ,			
4.3	StatusNotification (Unplug)	Include : connectorId (All Connector ID [0,1,2,3,...]), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
4.4	StatusNotification (Plug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
4.5	RemoteStartTransaction	Include : status (Accepted)			
4.6	StartTransaction	Include : connectorId , idTag , meterStart , timestamp			
4.7	StatusNotification	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
4.8	MeterValues (30 sec) *Energy.Active.Import.Register must be related to meterStart	Include : connectorId , transactionId , meterValue { timestamp , sampledValue {measurand (Voltage, Current.Import, Energy.Active.Import.Register, Power.Active.Import, SoC , Power.Offered), value , unit, context, location}}			
4.9	Reset (Soft)	Include : status			

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ลำดับ	Message	MEA OCPP Sever Required			Remark
		รายละเอียดการทดสอบ	Conn. ID 1	Conn. ID 2	
4.10	StopTransaction	Include : idTag, meterStop, timestamp, transactionId			
4.11	StatusNotification (Plug) (If any)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode, info, status, timestamp, vendorId, vendorErrorCode			
4.12	StatusNotification (Unplug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode, info, status, timestamp, vendorId, vendorErrorCode			
4.13	StatusNotification (Plug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode, info, status, timestamp, vendorId, vendorErrorCode			
4.14	RemoteStartTransaction	Include : status (Accepted)			
4.15	StartTransaction	Include : connectorId, idTag, meterStart, timestamp			
4.16	StatusNotification	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode, info, status, timestamp, vendorId, vendorErrorCode			
4.17	MeterValues (30 sec) *Energy.Active.Import.Register must be related to meterStart	Include : connectorId, transactionId, meterValue {timestamp, sampledValue {measurand (Voltage, Current.Import, Energy.Active.Import.Register, Power.Active.Import, SoC , Power.Offered), value, unit, context, location}}			
4.18	Reset (Hard)	Include : status			
4.19	StopTransaction	Include : idTag, meterStop, timestamp, transactionId			
4.20	StatusNotification (Plug) (If any)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode, info, status, timestamp, vendorId, vendorErrorCode			
4.21	StatusNotification (Unplug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode, info, status, timestamp, vendorId, vendorErrorCode			
5	การตรวจสอบการทำงานการจอง				
5.1	StatusNotification (Unplug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode, info, status, timestamp, vendorId, vendorErrorCode			
5.2	ReserveNow	Include : status			
5.3	StatusNotification (Unplug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode, info, status, timestamp, vendorId, vendorErrorCode			
5.4	CancelReservation	Include : status			

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ลำดับ	Message	MEA OCPP Sever Required			Remark
		รายละเอียดการทดสอบ	Conn. ID 1	Conn. ID 2	
5.5	StatusNotification (Unplug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
5.6	ReserveNow (1 min)	Include : status			
5.7	StatusNotification (Unplug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
5.8	StatusNotification (Unplug) (Timeout)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
5.9	ReserveNow	Include : status			
5.10	StatusNotification (Unplug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
5.11	StatusNotification (Plug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
5.12	RemoteStartTransaction	Include : status (Accepted)			
5.13	StartTransaction	Include : connectorId , idTag , meterStart , timestamp , reservationId			
5.14	StatusNotification	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
5.15	MeterValues (30 sec) *Energy.Active.Import.Register must be related to meterStart	Include : connectorId , transactionId , meterValue { timestamp , sampledValue { measurand (Voltage, Current.Import, Energy.Active.Import.Register, Power.Active.Import, SoC , Power.Offered), value , unit , context , location }}			
5.16	RemoteStopTransaction	Include : status			
5.17	StopTransaction	Include : idTag , meterStop , timestamp , transactionId			
5.18	StatusNotification (Plug) (If any)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
5.19	StatusNotification (Unplug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			

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ลำดับ	Message	MEA OCPP Sever Required			Remark
		รายละเอียดการทดสอบ	Conn. ID 1	Conn. ID 2	
6	การตรวจสอบการ Set Charging Profile				
6.1	StatusNotification (Unplug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode, info, status, timestamp, vendorId, vendorErrorCode			
6.2	StatusNotification (Plug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode, info, status, timestamp, vendorId, vendorErrorCode			
6.3	Authorize (Card)	Include : IdTag (Accepted)			
6.4	StartTransaction	Include : connectorId, idTag, meterStart, timestamp			
6.5	StatusNotification	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode, info, status, timestamp, vendorId, vendorErrorCode			
6.6	MeterValues (30 sec) *Energy.Active.Import.Register must be related to meterStart	Include : connectorId, transactionId, meterValue {timestamp, sampledValue {measurand (Voltage, Current.Import, Energy.Active.Import.Register, Power.Active.Import, SoC , Power.Offered), value, unit, context, location}}			
6.7	SetChargingProfile (chargingProfilePurpose: TxProfile, chargingRateUnit: W)	Include : status			
6.8	MeterValues (30 sec) *Energy.Active.Import.Register must be related to meterStart	Include : connectorId, transactionId, meterValue {timestamp, sampledValue {measurand (Voltage, Current.Import, Energy.Active.Import.Register, Power.Active.Import, SoC , Power.Offered), value, unit, context, location}}			
6.9	SetChargingProfile (chargingProfilePurpose: TxProfile, chargingRateUnit: W)	Include : status			
6.10	MeterValues (30 sec) *Energy.Active.Import.Register must be related to meterStart	Include : connectorId, transactionId, meterValue {timestamp, sampledValue {measurand (Voltage, Current.Import, Energy.Active.Import.Register, Power.Active.Import, SoC , Power.Offered), value, unit, context, location}}			
6.11	StopTransaction (Card)	Include : idTag, meterStop, timestamp, transactionId			
6.12	StatusNotification (Plug) (If any)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode, info, status, timestamp, vendorId, vendorErrorCode			

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ลำดับ	Message	MEA OCPP Sever Required			Remark
		รายละเอียดการทดสอบ	Conn. ID 1	Conn. ID 2	
6.13	StatusNotification (Unplug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
6.14	StatusNotification (Plug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
6.15	RemoteStartTransaction	Include : status (Accepted)			
6.16	StartTransaction	Include : connectorId , idTag , meterStart , timestamp			
6.17	StatusNotification	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
6.18	MeterValues (30 sec) *Energy.Active.Import.Register must be related to meterStart	Include : connectorId , transactionId , meterValue { timestamp , sampledValue {measurand (Voltage, Current.Import, Energy.Active.Import.Register, Power.Active.Import, SoC , Power.Offered), value , unit, context, location}}			
6.19	SetChargingProfile (chargingProfilePurpose: TxProfile, chargingRateUnit: W)	Include : status			
6.20	MeterValues (30 sec) *Energy.Active.Import.Register must be related to meterStart	Include : connectorId , transactionId , meterValue { timestamp , sampledValue {measurand (Voltage, Current.Import, Energy.Active.Import.Register, Power.Active.Import, SoC , Power.Offered), value , unit, context, location}}			
6.21	SetChargingProfile (chargingProfilePurpose: TxProfile, chargingRateUnit: W)	Include : status			
6.22	MeterValues (30 sec) *Energy.Active.Import.Register must be related to meterStart	Include : connectorId , transactionId , meterValue { timestamp , sampledValue {measurand (Voltage, Current.Import, Energy.Active.Import.Register, Power.Active.Import, SoC , Power.Offered), value , unit, context, location}}			
6.23	RemoteStopTransaction	Include : status			
6.24	StopTransaction	Include : idTag , meterStop , timestamp , transactionId			
6.25	StatusNotification (Plug) (If any)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			

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ลำดับ	Message	MEA OCPP Sever Required			Remark
		รายละเอียดการทดสอบ	Conn. ID 1	Conn. ID 2	
6.26	StatusNotification (Unplug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode, info, status, timestamp, vendorId, vendorErrorCode			
7	การตรวจสอบการทำงานรูปแบบผิดปกติ				
7.1	การ Remote Start แบบ Unplug				
7.1.1	StatusNotification (Unplug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode, info, status, timestamp, vendorId, vendorErrorCode			
7.1.2	RemoteStartTransaction	Include : status (Rejected)			
7.2	การ Remote Start Connector ID เดียวกันในเวลาใกล้เคียงกัน				
7.2.1	StatusNotification (Unplug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode, info, status, timestamp, vendorId, vendorErrorCode			
7.2.2	StatusNotification (Plug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode, info, status, timestamp, vendorId, vendorErrorCode			
7.2.3	RemoteStartTransaction	Include : status (Accepted)			
7.2.4	RemoteStartTransaction (same Connector ID within 10 sec)	Include : status (Rejected)			
7.2.5	StartTransaction	Include : connectorId, idTag, meterStart, timestamp			
7.2.6	StatusNotification	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode, info, status, timestamp, vendorId, vendorErrorCode			
7.2.7	MeterValues (30 sec) *Energy.Active.Import.Register must be related to meterStart	Include : connectorId, transactionId, meterValue {timestamp, sampledValue {measurand (Voltage, Current.Import, Energy.Active.Import.Register, Power.Active.Import, SoC , Power.Offered), value, unit, context, location}}			
7.2.8	RemoteStopTransaction	Include : status			
7.2.9	StopTransaction	Include : idTag, meterStop, timestamp, transactionId			
7.2.10	StatusNotification (Plug) (If any)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode, info, status, timestamp, vendorId, vendorErrorCode			
7.2.11	StatusNotification (Unplug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode, info, status, timestamp, vendorId, vendorErrorCode			

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ลำดับ	Message	MEA OCPP Sever Required			Remark
		รายละเอียดการทดสอบ	Conn. ID 1	Conn. ID 2	
7.3	การใช้ Card สลับใบ				
7.3.1	StatusNotification (Unplug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode, info, status, timestamp, vendorId, vendorErrorCode			
7.3.2	StatusNotification (Plug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode, info, status, timestamp, vendorId, vendorErrorCode			
7.3.3	Authorize (Invalid)	Include : IdTag			
7.3.4	Authorize (Accepted)	Include : IdTag			
7.3.5	StartTransaction	Include : connectorId, idTag, meterStart, timestamp			
7.3.6	StatusNotification	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode, info, status, timestamp, vendorId, vendorErrorCode			
7.3.7	MeterValues (30 sec) *Energy.Active.Import.Register must be related to meterStart	Include : connectorId, transactionId, meterValue {timestamp, sampledValue {measurand (Voltage, Current.Import, Energy.Active.Import.Register, Power.Active.Import, SoC , Power.Offered), value, unit, context, location}}			
7.3.8	Authorize (Invalid)	Include : IdTag			
7.3.9	StopTransaction (Card)	Include : idTag, meterStop, timestamp, transactionId			
7.3.10	StatusNotification (Plug) (If any)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode, info, status, timestamp, vendorId, vendorErrorCode			
7.3.11	StatusNotification (Unplug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode, info, status, timestamp, vendorId, vendorErrorCode			
7.4	การหยุดชาร์จด้วยวิธีดับปุ่ม Emergency				
7.4.1	StatusNotification (Unplug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode, info, status, timestamp, vendorId, vendorErrorCode			
7.4.2	StatusNotification (Plug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode, info, status, timestamp, vendorId, vendorErrorCode			
7.4.3	RemoteStartTransaction	Include : status (Accepted)			
7.4.4	StartTransaction	Include : connectorId, idTag, meterStart, timestamp			

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ลำดับ	Message	MEA OCPP Sever Required			Remark
		รายละเอียดการทดสอบ	Conn. ID 1	Conn. ID 2	
7.4.5	StatusNotification	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
7.4.6	MeterValues (30 sec) *Energy.Active.Import.Register must be related to meterStart	Include : connectorId , transactionId , meterValue { timestamp , sampledValue { measurand (Voltage, Current.Import, Energy.Active.Import.Register, Power.Active.Import, SoC , Power.Offered), value , unit , context , location }}			
7.4.7	StopTransaction (ตบปุ่ม Emergency)	Include : idTag , meterStop , timestamp , transactionId			
7.4.8	StatusNotification (Plug) *info must show the cause of the error	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
7.4.9	StatusNotification (Unplug) (If any)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
7.4.10	StatusNotification (Unplug) (คายนปุ่ม Emergency)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
7.5	การเปิดประตูเครื่องชาร์จ				
7.5.1	StatusNotification (Unplug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
7.5.2	StatusNotification (Open door) *info must show the cause of the error	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
7.5.3	StatusNotification (Close door)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
7.6	การทดสอบกรณีเกิดไฟฟ้าดับ				
7.6.1	StatusNotification (Unplug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
7.6.2	StatusNotification (Plug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
7.6.3	RemoteStartTransaction	Include : status (Accepted)			
7.6.4	StartTransaction	Include : connectorId , idTag , meterStart , timestamp			
7.6.5	StatusNotification	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			

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ลำดับ	Message	MEA OCPP Sever Required			Remark
		รายละเอียดการทดสอบ	Conn. ID 1	Conn. ID 2	
7.6.6	MeterValues (30 sec) *Energy.Active.Import.Register must be related to meterStart	Include : connectorId , transactionId, meterValue {timestamp, sampledValue {measurand (Voltage, Current.Import, Energy.Active.Import.Register, Power.Active.Import, SoC , Power.Offered), value , unit, context, location}}			
7.6.7	StopTransaction (ตัดไฟฟ้าเครื่องชาร์จ)	Include : idTag, meterStop , timestamp, transactionId			
7.6.8	BootNotification	Include : chargePointModel , chargePointVendor , firmwareVersion,			
7.6.9	StatusNotification (Plug)	Include : connectorId (All Connector ID [0,1,2,3,...]), errorCode , info, status , timestamp, vendorId, vendorErrorCode			
7.6.10	StatusNotification (Unplug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info, status , timestamp, vendorId, vendorErrorCode			
7.7	การทดสอบบัตรที่ลงทะเบียนไว้ใน LocalList				
7.7.1	ปลดสาย LAN				
7.7.2	เริ่มชาร์จโดย card ใน localAuthorizationList	ชาร์จได้สำเร็จ			
7.7.3	หยุดชาร์จโดย card ใน localAuthorizationList	หยุดชาร์จ			
8	การทดสอบชาร์จพร้อมกัน 2 หัว				
8.1	การ Remote Start พร้อมกัน 2 หัว				
8.1.1	StatusNotification (Unplug)	Include : connectorId (All Connector ID [0,1,2,3,...]), errorCode , info, status , timestamp, vendorId, vendorErrorCode			
8.1.2	StatusNotification (connectorId 1&2: Plug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info, status , timestamp, vendorId, vendorErrorCode			
8.1.3	RemoteStartTransaction (connectorId 1)	Include : status (Accepted)			
8.1.4	RemoteStartTransaction (connectorId 2) *within 10 sec after RemoteStart connectorId 1	Include : status (Accepted or Rejected) *Must be related to the behavior of the Charger - if Accepted, charger must StartTransaction within 90 sec - If Rejected, server must RemoteStartTransaction again			
8.1.5	StartTransaction (connectorId 1)	Include : connectorId , idTag, meterStart, timestamp			
8.1.6	StatusNotification	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info, status , timestamp, vendorId, vendorErrorCode			
8.1.7	StartTransaction (connectorId 2)	Include : connectorId , idTag, meterStart, timestamp			

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ลำดับ	Message	MEA OCPP Sever Required			Remark
		รายละเอียดการทดสอบ	Conn. ID 1	Conn. ID 2	
8.1.8	StatusNotification	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
8.1.9	MeterValues (30 sec) (connectorId 1) *Energy.Active.Import.Register must be related to meterStart	Include : connectorId , transactionId , meterValue { timestamp , sampledValue {measurand (Voltage, Current.Import, Energy.Active.Import.Register, Power.Active.Import, SoC , Power.Offered), value , unit , context , location }}			
8.1.10	MeterValues (30 sec) (connectorId 2) *Energy.Active.Import.Register must be related to meterStart	Include : connectorId , transactionId , meterValue { timestamp , sampledValue {measurand (Voltage, Current.Import, Energy.Active.Import.Register, Power.Active.Import, SoC , Power.Offered), value , unit , context , location }}			
8.1.11	RemoteStopTransaction (connectorId 1)	Include : status			
8.1.12	StopTransaction	Include : idTag , meterStop , timestamp , transactionId			
8.1.13	StatusNotification (If any)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
8.1.14	StatusNotification (connectorId 1: Unplug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
8.1.15	RemoteStopTransaction (connectorId 2)	Include : status			
8.1.16	StopTransaction	Include : idTag , meterStop , timestamp , transactionId			
8.1.17	StatusNotification (If any)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
8.1.18	StatusNotification (connectorId 2: Unplug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
8.2	การหยุดชาร์จด้วยวิธีดับปุ่ม Emergency				
8.2.1	StatusNotification (Unplug)	Include : connectorId (All Connector ID [0,1,2,3,...]), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
8.2.2	StatusNotification (connectorId 1: Plug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
8.2.3	RemoteStartTransaction (connectorId 1)	Include : status (Accepted)			
8.2.4	StartTransaction	Include : connectorId , idTag , meterStart , timestamp			

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ลำดับ	Message	MEA OCPP Sever Required			Remark
		รายละเอียดการทดสอบ	Conn. ID 1	Conn. ID 2	
8.2.5	StatusNotification	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
8.2.6	MeterValues (30 sec) *Energy.Active.Import.Register must be related to meterStart	Include : connectorId , transactionId , meterValue { timestamp , sampledValue {measurand (Voltage, Current.Import, Energy.Active.Import.Register, Power.Active.Import, SoC , Power.Offered), value , unit , context , location }}			
8.2.7	StatusNotification (connectorId 2: Plug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
8.2.8	RemoteStartTransaction (connectorId 2)	Include : status (Accepted)			
8.2.9	StartTransaction	Include : connectorId , idTag , meterStart , timestamp			
8.2.10	StatusNotification	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
8.2.11	MeterValues (30 sec) *Energy.Active.Import.Register must be related to meterStart	Include : connectorId , transactionId , meterValue { timestamp , sampledValue {measurand (Voltage, Current.Import, Energy.Active.Import.Register, Power.Active.Import, SoC , Power.Offered), value , unit , context , location }}			
8.2.12	StopTransaction (ตบปุ่ม Emergency) (connectorId 1)	Include : idTag , meterStop , timestamp , transactionId			
8.2.13	StopTransaction (ตบปุ่ม Emergency) (connectorId 2)	Include : idTag , meterStop , timestamp , transactionId			
8.2.14	StatusNotification (connectorId 1&2: Plug) *info must show the cause of the error	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
8.2.15	StatusNotification (connectorId 1&2: Unplug) (If any)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
8.2.16	StatusNotification (connectorId 1&2: Unplug) (ค้ายปุ่ม Emergency)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
8.3	การทดสอบกรณีเกิดไฟฟ้าดับ				
8.3.1	StatusNotification (Unplug)	Include : connectorId (All Connector ID [0,1,2,3,...]), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
8.3.2	StatusNotification (connectorId 1: Plug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
8.3.3	RemoteStartTransaction (connectorId 1)	Include : status (Accepted)			
8.3.4	StartTransaction	Include : connectorId , idTag , meterStart , timestamp			

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ลำดับ	Message	MEA OCPP Sever Required			Remark
		รายละเอียดการทดสอบ	Conn. ID 1	Conn. ID 2	
8.3.5	StatusNotification	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
8.3.6	MeterValues (30 sec) *Energy.Active.Import.Register must be related to meterStart	Include : connectorId , transactionId , meterValue { timestamp , sampledValue { measurand (Voltage, Current.Import, Energy.Active.Import.Register, Power.Active.Import, SoC , Power.Offered), value , unit , context , location }}			
8.3.7	StatusNotification (connectorId 2: Plug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
8.3.8	RemoteStartTransaction (connectorId 2)	Include : status (Accepted)			
8.3.9	StartTransaction	Include : connectorId , idTag , meterStart , timestamp			
8.3.10	StatusNotification	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
8.3.11	MeterValues (30 sec) *Energy.Active.Import.Register must be related to meterStart	Include : connectorId , transactionId , meterValue { timestamp , sampledValue { measurand (Voltage, Current.Import, Energy.Active.Import.Register, Power.Active.Import, SoC , Power.Offered), value , unit , context , location }}			
8.3.12	StopTransaction (ไฟฟ้าดับ) (connectorId 1)	Include : idTag , meterStop , timestamp , transactionId			
8.3.13	StopTransaction (ไฟฟ้าดับ) (connectorId 2)	Include : idTag , meterStop , timestamp , transactionId			
8.3.14	BootNotification	Include : chargePointModel , chargePointVendor , firmwareVersion ,			
8.3.15	StatusNotification (connectorId 1&2: Plug)	Include : connectorId (All Connector ID [0,1,2,3,...]), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
8.3.16	StatusNotification (connectorId 1&2: Unplug)	Include : connectorId (All Connector ID [0,1,2,3,...] if status change), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
9	การตั้งค่าเครื่องชาร์จเป็นค่าที่ กฟน. กำหนด				
9.1	ChangeConfiguration (HeartbeatInterval: 60 min)	Include : status (Accepted)			
9.2	ChangeConfiguration (MeterValueSampleInterval: 1 min)	Include : status (Accepted)			
9.3	ChangeConfiguration (LocalAuthorizeOffline: False)	Include : status (Accepted)			

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ลำดับ	Message	MEA OCPP Sever Required			Remark
		รายละเอียดการทดสอบ	Conn. ID 1	Conn. ID 2	
10	สรุปผลการตรวจสอบ Messages				
10.1	Authorize	Include : IdTag			
10.2	BootNotification	Include : chargePointModel , chargePointVendor , firmwareVersion			
10.3	DataTransfer				*ขึ้นกับการพิจารณา
10.4	Heartbeat (Interval 60 min)	Include :			
10.5	MeterValues (Interval 1 min)	Include : connectorId , transactionId , meterValue { timestamp , sampledValue { measurand (Voltage, Current.Import, Energy.Active.Import.Register, Power.Active.Import, SoC , Power.Offered),			
10.6	RemoteStartTransaction	Include : status (Accepted or Rejected (if Unplug))			
10.7	RemoteStopTransaction	Include : status			
10.8	Reset	Include : status			
10.9	StartTransaction	Include : connectorId , idTag , meterStart , timestamp			
10.10	StatusNotification	Include : connectorId (All Connector ID [0,1,2,3,...]), errorCode , info , status , timestamp , vendorId , vendorErrorCode			
10.11	StopTransaction	Include : idTag , meterStop , timestamp , transactionId			
10.12	ReserveNow	Include : status			
10.13	CancelReservation	Include : status			
10.14	SetChargingProfile (chargingProfilePurpose: TxProfile, chargingRateUnit: W)	Include : status			
10.15	TriggerMessage	Include : status (Accepted)			
10.16	GetConfiguration	Include : configurationKey { HeartbeatInterval (int), LocalAuthorizeOffline (boolean), MeterValueSampleInterval (int), UnlockConnectorOnEVSideDisconnect (boolean)}			
10.17	ChangeConfiguration	Include : status (Accepted)			
10.18	ChangeAvailability	Include : status (Accepted)			
10.19	GetDiagnostics	Include : fileName			*ขึ้นกับการพิจารณา
10.20	UpdateFirmware	Include :			*ขึ้นกับการพิจารณา
10.21	SendLocalList (listVersion, updateType: Full , [localAuthorizationList])	Include : status (Accepted)			

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ลำดับ	Message	MEA OCPP Sever Required			Remark
		รายละเอียดการทดสอบ	Conn. ID 1	Conn. ID 2	
10.22	GetLocalListVersion	Include : listVersion			
10.23	clearCache	Include : status (Accepted)			
11	การตรวจสอบอื่นๆ				
11.1	ปลดสาย LAN	Handshaking ในเวลา นาที (< 1 min)			
11.2	การ Disconnect	ครั้ง / ชั่วโมง เวลา Handshaking ใช้เวลา นาที			